

Power Budget Example

Team Number:	209							
Project Name:	Smart Door Sensor							
Team Member Names:	Matthew Sanderson, Andrew Imoukhuede, Dylan Vierra-Guillermo, Bryce Weber							
Version:	1							
All Major Components	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit	
	74HC165 Shift Register	74HC165	+2 - 6V	1	60	60	mA	
	Curiosity Nano	PIC18F57Q43	+1.8 - 5.1V	1	500	500	mA	
	AOT2618L MOSFET	AOT2618L TO-220	-20V - +20V	1	1400	1400	mA	
+5V Power Rail	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit	
	74HC165 Shift Register	74HC165	+2 - 6V	1	60	60	mA	
	Curiosity Nano	PIC18F57Q43	+1.8 - 5.1V	1	500	500	mA	
	Subtotal					560	mA	
	Safety Margin					25%		
	Total Current Required on +5V Rail					700	mA	
	Regulator	AOT2618L MOSFET	AOT2618L TO-220	-20V - +20V	1	1400	1400	mA
Total Remaining Current Available on +5V Rail					700	mA		
External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit	
	Power Source 1 Selection	Plug-in Wall Supply	(full part number)	110VAC	+9V	3000	5000	mA
	Power Rails Connected to External Power Source 1	AOT2618L MOSFET	AOT2618L TO-220	-20V - +20V	1	1400	1400	mA
	Total Remaining Current Available on External Power Source 1					3600	mA	